CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Forth Semester of B. Tech. Examination (IT)

May 2012

MA 206 Probability, Statistics and Numerical Analysis

Date: 15/05/2012, Tuesday Time: 01:30am to 04:30pm Maximum Marks: 70

Instruction:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1(a) Discuss Pie-chart with at least one real life example. [04]

Q - 1(b) Write probability density function of following distributions: [03]
(i) Uniform (ii) Poisson (iii) Binomial.

Q - 2(a) Compute the Arithmetic Mean for the following data. [05]

Marks	0-10	10-20	20-30	30-40	40-50
No. of students	4	36	60	34	6

(b) Find the correlation coefficient between the sales and expenses of the following 10 firms using Karl Pearson's Product moment method. (Figures are in thousands) [05]

Firms	1	2	3	4	5	6	7	8	9	10
Sales	50	50	55	60	65	65	65	60	60	50
Expenses	11	13	14	16	16	15	15	14	13	13

(c) A card is Drawn at random from a pack of 52 cards. What is the probability that the card is a club or a Ace? [04]

OR

Q - 2(a) Two judges in a beauty contest rank the 12 contestants as follows [05]

X	1	2	3	4	5	6	7	8	9	10	11	12
Y	12	9	6	10	3	5	4	7	8	2	11	1

What degree of agreement is there between the judges? Use Spearman's Rank Correlation Method.

(b) Draw a Histogram and draw your conclusion for the data given below [05]

Marks	0-10	10-20	20-30	30-40	40-50
No. of students	4	36	60	34	6

(c) (i) An experiment consists of tossing two dice. What is the sample space of this experiment? [04]

(ii) An electronic controlling mechanism requires five identical memory chips. In how many ways can this mechanism be assembled by placing the 5 chips in the 5 positions within controller?

Q - 3(a) From the following data, obtain the regression line of x on y. [05]

x	1	2	3	4	5
y	27	33	15	24	21

(b) It is known that screws produced by a certain company will be defective with probability 0.01 independently of each other. The company sells the screws in packages of 10 and offers money - back guarantee that at most 1 of the 10 screws is defective. What proportion of packages sold must the company replace? By using Binomial distribution. [05]

(c) Find the Standard Deviation of 12, 16, 22, 13, 20, 14, 11, 14 and 20 . [04]

OR

Q - 3(a) From the following data, obtain the regression line of y on x [05]

x	1	2	3	4	5
y	27	33	15	24	21

(b) X follows a Poisson distribution with standard deviation 1.5, Find P $(X \geq 3)$ [05]

(c) A sample of 100 dry battery cells tested and found that average life of 12 hours and a standard deviation 3 hours. Assuming the data to be Normally distributed, what percentage of battery cells are expected to have life between 9 and 12 hours? [04]

SECTION - II

Q - 4 State whether the following statements are true or false: [07]

(a) percentage error is 100 multiple of relative error.

(b) $E\Delta = \Delta E$.

(c) Minimum number of subinterval required to apply Simpson's rules is 6.

(d) Runge-Kutta 2$^{\text{nd}}$ order method produce more accurate solution than Euler's Method.

(e) Regula falsi method is faster than bisection method.

(f) $(101011.1)_2 - (43.2)_8 = (0.3)_{10}$.

(g) Simpson's $\frac{1}{3}^{\text{rd}}$ rule determined by substituting $n = 3$ (no. of sub intervals) in Newton's-Cotes quadrature formula.

Q - 5(a) Using Newton's Interpolation formula find $f(1975)$ and $f(2005)$ from the following data: [05]

x	1971	1981	1991	2001	2011
$f(x)$	46	66	81	93	101

(b) Evaluate $\int_{0.4}^{1.6} e^{-x^2} dx$, by Trapezoidal and Simpson's 3/8 rule with $n =$ 6subintervals. [05]

(c) Using Newton-Raphson method, find a root of the function $f(x) = e^x - 3x^2$ to an accuracy of 5 digits. The root is known to lie between 0.5 and 1.0. Take the starting value of x as $x_0 = 1.0$. [04]

OR

Q - 5(a) Apply appropriate interpolation formula to find a polynomial $f(x)$ which passes through the points (0, 20), (1, 12), (3, 20) and (4, 24). Hence evaluate $f(2)$. [05]

(b) Determine the equation to the best fitting exponential curve of the form $y = ae^{bx}$ for the data given below: [05]

x	1	2	3	4	5
y	22	8	3	1	0.35

(c) Solve $y' = x - y^2$ with $y(0) = 0$, using Picard's method. [04]

Q - 6(a) Use the Runge-Kutta method of order four with $h = 0.2$ to obtain an approximation to $y(1.2)$ for the solution of $y' = 2y^2$, $y(1) = 1$. [05]

(b) Define the following terms with suitable example. [05]

(i) Inherent error (ii) Significant digits (iii) Truncation error (iv) Round-off error

(c) Determine the equation to the best fitting exponential curve of the form $y = ax + b$ for the data given below: [04]

x	-1	0	1	2
y	1	0	1	4

OR

Q - 6(a) Integrate by Gaussian quadrature ($n = 3$): $\int_1^2 \frac{1}{1+x^3} dx$. [05]

(b) Use the Bisection method to find a root of the equation $x^3 - 4x - 8.95 = 0$ accurate to two decimal places which lies in [2, 3]. [05]

(c) Give at least two source in which error exist. How to reduce those errors? Justify. [04]

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z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990